

Functional Dependency (FD) in DBMS

Complete Core Guide, Types, Armstrong's Axioms, Rules & GATE Reference

Functional Dependency (FD) is a foundational constraint in relational database design. It defines how attributes within a relation relate to and uniquely determine one another. Mastering FDs is essential for:

- Finding **Candidate Keys** and **Superkeys**
- Calculating **Attribute Closures** (X^+)
- Database **Normalization** (2NF, 3NF, and BCNF)
- Solving university and competitive exam questions (GATE, UGC NET, etc.)

1. What is Functional Dependency?

A Functional Dependency describes a relationship between attributes in a relation. It answers the question: "If we know the value of attribute set X , can we uniquely determine the value of attribute set Y ?"

$X \rightarrow Y$ (Read as: " X determines Y " or " Y is functionally dependent on X ")

2. Determinant and Dependent

In any functional dependency $X \rightarrow Y$:

- **Determinant (X):** The attribute (or set of attributes) on the Left-Hand Side (LHS) that determines another attribute.
- **Dependent (Y):** The attribute (or set of attributes) on the Right-Hand Side (RHS) whose value is determined by X .

X (Determinant) $\longrightarrow$ Y (Dependent)

Real-World Example: $SID \rightarrow S_Name$ means "Student ID uniquely determines Student Name". If you know a student's SID , there is exactly one student name associated with it.

3. Understanding FD with Table Data

Consider the following student relation table:

SID	S_Name	Validity Check for $SID \rightarrow S_Name$
1	Ranjit	$SID = 1 \rightarrow \text{Ranjit}$ (Valid)
2	Ranjit	$SID = 2 \rightarrow \text{Ranjit}$ (Valid: Two students can share the same name)
3	Varun	$SID = 3 \rightarrow \text{Varun}$ (Valid)

What Violates a Functional Dependency?

Suppose the relation contained the following conflicting rows:

SID	S_Name	Status
1	Ranjit	$SID = 1$ maps to "Ranjit"
1	Varun	VIOLATION! $SID = 1$ maps to "Varun"

Since $SID = 1$ maps to two different names ("Ranjit" and "Varun"), the dependency $SID \rightarrow S_Name$ is **violated** and does not hold for this relation.

4. The Golden Rule of Functional Dependency ★

Golden Rule: For every specific value of X , there must be **exactly one** corresponding value of Y .

- Different X values can determine the same Y value (e.g., $SID\ 1 \rightarrow Ranjit$, $SID\ 2 \rightarrow Ranjit$ is VALID).
- The same X value **CANNOT** determine different Y values (e.g., $SID\ 1 \rightarrow Ranjit$, $SID\ 1 \rightarrow Varun$ is INVALID).

🧠 *Memory Trick:* **Same $X \rightarrow$ Same Y**

5. Types of Functional Dependencies

A. Trivial Functional Dependency

A functional dependency $X \rightarrow Y$ is **trivial** if:

$$Y \subseteq X \quad (\text{RHS is a subset of LHS})$$

Examples:

- $SID \rightarrow SID$ (Trivial because $\{SID\} \subseteq \{SID\}$)
- $\{SID, SName\} \rightarrow SID$ (Trivial because $\{SID\} \subseteq \{SID, SName\}$)
- $AB \rightarrow A$ (Trivial because $\{A\} \subseteq \{A, B\}$)

Key Property: Trivial dependencies are *always true* for any relational instance and do not require testing data.

B. Non-Trivial Functional Dependency

A functional dependency $X \rightarrow Y$ is **non-trivial** if:

$$Y \not\subseteq X \quad (\text{RHS is NOT completely contained in LHS})$$

Examples: $SID \rightarrow SName$, $EmployeeID \rightarrow Location$, $AB \rightarrow BC$ (since $\{B, C\} \not\subseteq \{A, B\}$).

C. Completely Non-Trivial Functional Dependency

A functional dependency $X \rightarrow Y$ is **completely non-trivial** if:

$$X \cap Y = \emptyset \quad (\text{LHS and RHS share NO common attributes})$$

Example: $SID \rightarrow SName$ (assuming SID and $SName$ are distinct attributes).

FD Category	Formal Condition	Example
Trivial	$Y \subseteq X$	$AB \rightarrow A$
Non-Trivial	$Y \not\subseteq X$	$AB \rightarrow BC$
Completely Non-Trivial	$X \cap Y = \emptyset$	$A \rightarrow B$

6. Armstrong's Axioms (Core Inference Rules)

Armstrong's Axioms are a set of fundamental rules used to infer all valid functional dependencies from a given set of FDs.

Primary Axioms:

1. Reflexivity Rule:

If $Y \subseteq X$, then $X \rightarrow Y$.

Example: $ABC \rightarrow BC$ (Yields trivial FDs).

2. Augmentation Rule:

If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any attribute set Z .

Example: Given $SID \rightarrow SName$, adding **Phone** to both sides gives $\{SID, Phone\} \rightarrow \{SName, Phone\}$.

 *Memory Trick:* Add the exact same attribute set to both sides.

3. Transitivity Rule ★★☆☆:

If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$.

Example: Given $SID \rightarrow SName$ and $SName \rightarrow City$, we infer $SID \rightarrow City$.

 *Memory Trick:* X reaches Z through intermediate Y .

7. Secondary / Derived Rules

4. Union Rule:

If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$.

Example: If $SID \rightarrow SName$ and $SID \rightarrow Age$, then $SID \rightarrow \{SName, Age\}$.

5. Decomposition Rule ★:

If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$.

Example: If $SID \rightarrow \{SName, Age\}$, we can decompose into $SID \rightarrow SName$ and $SID \rightarrow Age$.

⚠ CRITICAL WARNING: NEVER Decompose the Left-Hand Side (LHS)!

If $XY \rightarrow Z$, you **CANNOT** conclude $X \rightarrow Z$ or $Y \rightarrow Z$.

Real-World Example: $\{StudentID, CourseID\} \rightarrow Grade$ means StudentID and CourseID *together* determine Grade. Neither StudentID alone nor CourseID alone determines the Grade.

6. Pseudo-Transitivity Rule:

If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$.

Example: If $A \rightarrow B$ and $CB \rightarrow D$, then $CA \rightarrow D$ (Substitute A for B in second dependency).

7. Composition Rule:

If $X \rightarrow Y$ and $Z \rightarrow W$, then $XZ \rightarrow YW$.

Example: If $A \rightarrow B$ and $C \rightarrow D$, then $AC \rightarrow BD$.

8. Connection: FD Rules & Attribute Closure

Functional Dependency rules directly enable the calculation of **Attribute Closure** (X^+) to find Superkeys and Candidate Keys:

Set of Functional Dependencies (FDs) $\rightarrow$ Apply Armstrong Rules $\rightarrow$ Calculate Closure (X^+) $\rightarrow$ Determines all attributes ($X^+ = R$)? $\rightarrow$ Superkey $\rightarrow$ Minimal? $\rightarrow$ Candidate Key

Example: Given $R(A, B, C, D)$ with FDs $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$:

- By Transitivity: $A \rightarrow B$, $A \rightarrow C$, and $A \rightarrow D$.
- By Union Rule: $A \rightarrow ABCD$.
- Therefore, $A^+ = \{A, B, C, D\} = R \rightarrow A$ is a **Candidate Key**.

9. Master Summary Cheat Sheet

Property / Rule	Formal Definition	Key Exam Takeaway
Functional Dependency	$X \rightarrow Y$	Each X value maps to exactly one Y value
Trivial FD	$Y \subseteq X$	Always holds true automatically
Reflexivity	If $Y \subseteq X \Rightarrow X \rightarrow Y$	Base axiom for subset determination
Augmentation	If $X \rightarrow Y \Rightarrow XZ \rightarrow YZ$	Add same attributes to both sides
Transitivity	If $X \rightarrow Y$ & $Y \rightarrow Z \Rightarrow X \rightarrow Z$	Chain connection rule
Union	If $X \rightarrow Y$ & $X \rightarrow Z \Rightarrow X \rightarrow YZ$	Combine common LHS rules
Decomposition	If $X \rightarrow YZ \Rightarrow X \rightarrow Y$ & $X \rightarrow Z$	RHS can split; LHS NEVER splits!
Pseudo-transitivity	If $X \rightarrow Y$ & $WY \rightarrow Z \Rightarrow WX \rightarrow Z$	Substitute X for Y in compound LHS
Composition	If $X \rightarrow Y$ & $Z \rightarrow W \Rightarrow XZ \rightarrow YW$	Combine both LHS and RHS